

Target Selection with Avatars in Mixed Reality

Eric DeMarbre*

Robert J. Teather†

Carleton University School of Information Technology

ABSTRACT

This poster presents a Fitts' law experiment evaluating the effects of using an avatar in mixed and virtual reality selection tasks. The avatar had little to no impact on efficiency and surprisingly lowered overall accuracy in both the 2D plane and depth. However, the avatar also reduced variability in depth selection. Avatar design and specific MR hardware parameters may significantly impact efficiency, especially compared to VR devices.

Index Terms: Human-centered computer – Virtual Reality; Human-centered computer – Human Computer Interaction (HCI); Human-centered computer – Mixed Reality;

1 INTRODUCTION

Video see-through mixed reality (MR) capabilities are increasingly prevalent on modern head-mounted displays like the Meta Quest 3. Such systems are susceptible to occlusion problems between opaque virtual objects and real-world objects. Mismatched parallax, where virtual and real objects don't move or rotate as expected, may impact the user's depth judgment of virtual objects. More critically, the user's real hands are occluded by virtual objects when passing behind them, despite the virtual object being farther away than the hands. Occlusion is the most powerful depth cue humans rely on for depth judgment, so occlusion mismatches impact a user's ability to interact with virtual objects in depth. Virtual representations of the controllers can partially offset this. Still, as hand-tracking is also becoming increasingly prevalent, the presence of VR controllers can no longer be assumed on such devices.

Adding a virtual avatar may solve these problems. The avatar's virtual body can replace a user's and replicate their movement by tracking the head and hand positions. Avatars are known to improve presence and body possession in VEs, but there are mixed results in their impact on user performance [3]. We conducted an experiment evaluating the effect of avatars in mixed reality on user performance using a 2D Fitts' law target selection task displayed on a Meta *Quest Pro* VR head-mounted display (HMD). Participants completed selection tasks in MR and VR environments, both with and without an avatar mimicking their hand and arm motions.

2 RELATED WORK

To perceive depth, humans rely on numerous depth cues, including binocular disparity, texture gradient, and motion parallax [2]. Previous work on VR avatars [6] has shown that sufficiently detailed avatars can improve performance over a less-detailed cartoon avatar, especially with texture detail. There are

comparatively few studies on the effect of avatars on user performance in MR, but augmented reality (AR) has been used in several similar Fitts' law experiments [1,5]. Mason et al. report some benefits to target selection when having avatars in AR.

Our study employs a 3D variant of the ISO 9241-9 standard reciprocal selection task [7], so we briefly describe this and Fitts' law, upon which the standard is based. Fitts' law models the relationship between selection time (MT) and selection task difficulty (ID , in bits) as:

$$MT = a + b \times ID \quad (1)$$

where the index of difficulty (ID) is based on target distance (amplitude, A) and size (width, W) as

$$ID = \log_2 \left(\frac{A}{W} + 1 \right) \quad (2)$$

While initially developed for 1D motions, later work [4] adapted the task to multiple directions using targets arranged in a circular formation. The ISO standard recommends using the *effective index of difficulty* (ID_e) through post-experimental adjustment to width and amplitude [7]. Effective width (W_e) adjusts the experimental error rate to 4%, while effective amplitude (A_e) adjusts the target movement distance to the average of the actual movement distance. ID_e is given as:

$$ID_e = \log_2 \left(\frac{A_e}{W_e} + 1 \right), \text{ where } W_e = 4.113 \times SD_x \quad (3)$$

SD_x is the standard deviation of selection coordinates along the task axis, accounting for selection coordinate variability. ID_e and the recorded mean selection time are used to derive throughput (TP , in bits per second) as

$$TP = \left(\frac{ID_e}{MT} \right) \quad (4)$$

The primary advantage of throughput is that it incorporates speed and accuracy (through W_e) into a single measure, representing the effectiveness of the selection technique. Through numerous studies, mouse selection throughput is approximately 4 bps [4,8], but it has also been employed in evaluating 3D input devices and selection techniques [7].

3 EXPERIMENT

We recruited 21 participants (11 male, 8 female, 1 fluid, 1 transmasculine; mean age of 26.4 ± 7.21 years; 19 right-handed). All had stereo vision and could use Meta Quest Touch controllers.

We used a Meta Quest Pro VR and two touch controllers for the selection tasks and other interactions in the VEs. We developed two VEs in Unity 3D, a fully VR and MR environment. The VR environment was an approximately 1:1 scale virtual replica of the lab where the experiment occurred. See Figure 3 left. The MR environment featured only the virtual targets and no other virtual objects (Figure 2 right), with the passthrough cameras enabled so that participants could see the regular environment plus virtual targets. Virtual targets are seen in Figure 2.

* ericdemarbre@cmail.carleton.ca

† robteather@cunet.carleton.ca

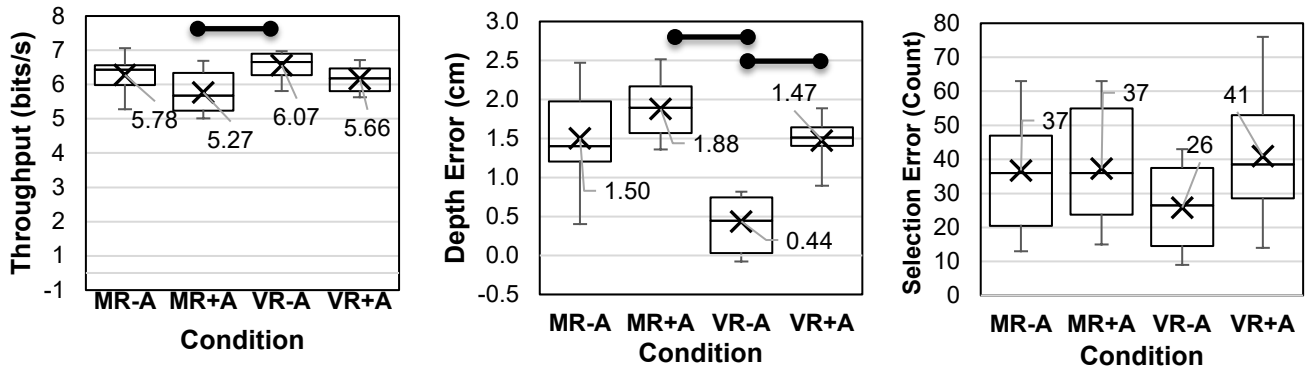


Figure 2. Experiment results by dependent variable. (Left) Throughput. (Centre) Depth error. (Right) Error count. Black bars (●) between box plots indicate pairwise significant differences per Tukey's HSD posthoc test ($p < .05$)

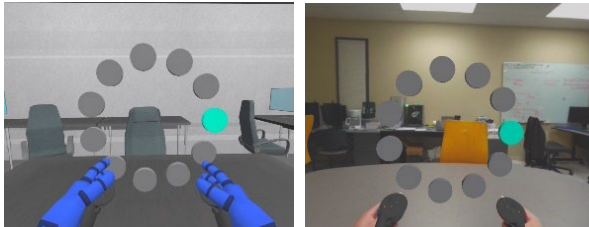


Figure 1. Selection Targets in (Left) VR and (Right) MR.

The avatar condition featured a simple, low-polygon model with an inverse-kinematic rig attached (Figure 3 right). This rig followed the hand and head movements of the participant and simulated the movements of the attached arms. The model could be activated and deactivated on demand.



Figure 3. (Left) Virtual office environment and (Right) rigged avatar.

The task involved selecting each of the 11 targets seen in Figure 2, each highlighting in turn to indicate it should be selected. The software recorded time and selection accuracy for each target. Participants were instructed to select the blue target as quickly and accurately as possible by touching the controller to the target and pulling the trigger button. The participants repeated this process until all 6 ID combinations were completed for each condition.

The study employed a $2 \times 2 \times 3 \times 2 \times 2$ within-subjects design, with two primary independent variables, XR mode (VR, MR) and avatar (on, off), plus three target distances (0.5 m, 0.6 m, 0.7 m) and two target widths: 0.1 m and 0.06 m) to create a range of 6 IDs, that were repeated a total of two times each. We recorded 11088 selections, with 21 participants selecting 11 targets over 12 IDs and 4 conditions ($21 \times 11 \times 12 \times 4 = 11088$). Dependent variables included throughput, calculated via equation 5, selection errors (count of targets missed), and depth error (inaccuracy in depth, in cm).

4 RESULTS & DISCUSSION

Results are summarized in Figure 1. The main effect for both XR mode ($F_{1,11} = 23.9$, $p < .0005$) and avatar ($F_{1,11} = 67.2$, $p < .001$) were significant for throughput. The interaction effect between XR mode and avatar was significant for depth error ($F_{1,11} = 22.2$, $p < .001$), as were both the main effect for XR mode ($F_{1,11} = 100.7$, $p < .0001$), and avatar ($F_{1,11} = 87.5$, $p < .0001$). Finally, there was a significant interaction between XR mode and avatar for selection error ($F_{1,11} = 18.2$, $p < 0.005$) and a significant main effect for avatar mode on selection error ($F_{1,11} = 16.3$, $p < .005$). Post hoc testing failed to detect a pairwise difference, however.

Throughput was generally lower in MR than in VR; MR+A was significantly lower. XR mode thus appears to yield an effect on selection performance. Contrary to our expectations, selection accuracy was negatively impacted by the avatar. Depth error was substantially and significantly lower with VR-A compared to all other conditions. This suggests that the avatar did not help participants determine target depth. This is further supported by the substantially lower selection error with VR-A. However, in both cases, the +A condition reduced the variance of the depth error.

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5 CONCLUSION AND FUTURE WORK

While the experiment results are mixed, avatars may still positively impact user experience in assisting with the accuracy of selection tasks. Future work will explore alternative avatar designs, which may have greater impact than an avatar's mere presence or absence.

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